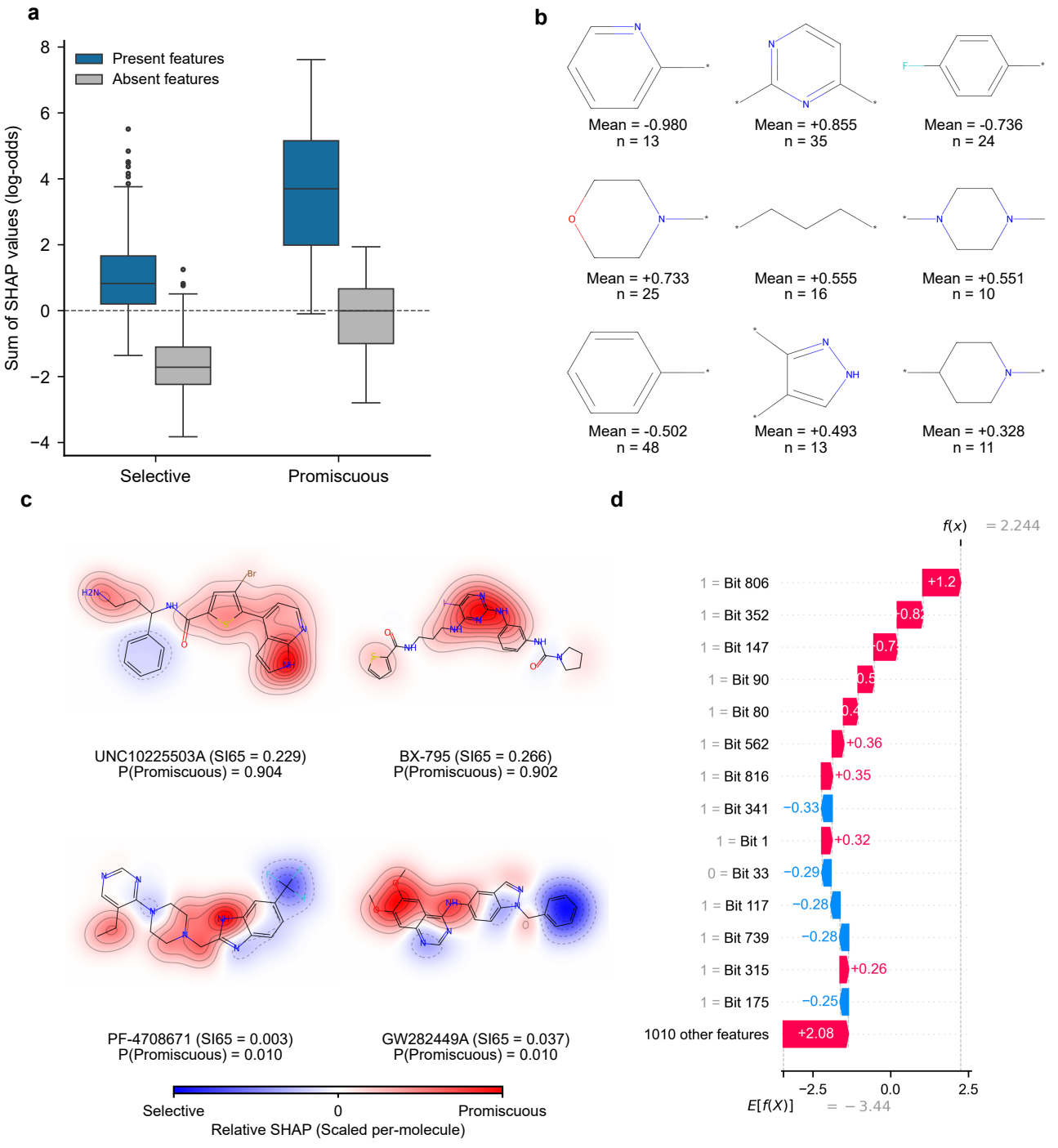

a
Present features
Absent features
Sum of SHAP values (log-odds)
8
6
4
2
0
−2
−4
Selective
Promiscuous
b
Mean = -0.980
n = 13
Mean = +0.855
n = 35
Mean = -0.736
n = 24
Mean = +0.733
n = 25
Mean = +0.555
n = 16
Mean = +0.551
n = 10
Mean = -0.502
n = 48
Mean = +0.493
n = 13
Mean = +0.328
n = 11
c
UNC10225503A (SI65 = 0.229)
P(Promiscuous) = 0.904
BX-795 (SI65 = 0.266)
P(Promiscuous) = 0.902
PF-4708671 (SI65 = 0.003)
P(Promiscuous) = 0.010
GW282449A (SI65 = 0.037)
P(Promiscuous) = 0.010
Selective
0
Promiscuous
Relative SHAP (Scaled per-molecule)
d
f(x) = 2.244
1 = Bit 806 +1.2
1 = Bit 352
1 = Bit 147
1 = Bit 90
1 = Bit 80
1 = Bit 562 +0.36
1 = Bit 816 +0.35
1 = Bit 341 −0.33
1 = Bit 1 +0.32
0 = Bit 33 −0.29
1 = Bit 117 −0.28
1 = Bit 739 −0.28
1 = Bit 315 +0.26
1 = Bit 175 −0.25
1010 other features +2.08
−2.5
0.0
2.5
E[f(X)] = −3.44